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Ground following snow plow attachment

Abstract

An attachment assembly for attaching a snow plow blade to a vehicle includes a pair of interface brackets, a pair of articulated intermediate structures mounted to the interface brackets, a transversal member joining the intermediate structures, and a trigger assembly. The trigger assembly includes a static member with a pivoting attachment point for attachment to the snow plow blade and top and bottom attachment points for releasable attachment to the intermediate structure, a pin assembly, and a biasing means attached to the pin assembly and to the static member. The trigger assembly is biased toward a default position in which the pin member and the static member are at their closest. When applying a torque on the snow plow blade above a trigger threshold, the snow plow blade rotates from the default position to a triggered position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority from U.S. provisional patent application 62/968,519 filed Jan. 31, 2020, the specification of which is hereby incorporated herein by reference in its entirety.

BACKGROUND

(a) Field

(1) The present invention generally relates to blades for snow plows, and more particularly relates to snow plow blades assemblies and attachment assemblies for snow plow blades that comprise mechanisms capable to adapt to the unevenness of the ground.

(b) Related Prior Art

(2) Snow plows are both well-known and each has a relatively long snow plow blade assembly which extends generally laterally of the surface being worked on. Some of them comprise sidewalls that retain the snow sideways, allowing to controllably displace more snow before having an overflow flowing out sideways of the snow plow blade assembly.

(3) Further, the ground the snow plow tries to clean is by nature not always even. In order to optimize the process of grading the ground with the snow plow, the snow plow blade should be able to adapt to the changes of slope of the ground, not only in the direction of the grading but also transversally to the grading.

(4) There is therefore a need for a solution for the snow plow blade assembly to automatically adapt to the unevenness of the ground while maintaining the main characteristics of the snow plow comprising the capacity to lift the snow plow blade assembly, the stability of the snow plow blade assembly both in grading operation and when lifted above the ground, and the other controls regarding the snow plow blade assembly,

(5) There is further a need for a solution for the snow blade assembly to decrease the impact resulting from hitting an obstacle, more specifically undergone by the snow plow blade assembly itself and by the vehicle on which is mounted the snow plow blade assembly, thereby decreasing premature wear that may result from these impacts.

(6) Therefore, there is a need for an improved mechanism for snow plow blade assemblies that better respond to these objectives.

SUMMARY

(7) According to an embodiment, there is provided an attachment assembly for attaching a snow plow blade to a vehicle, comprising: an interface bracket adapted for releasable connection with the vehicle; an intermediate structure mounted to the interface bracket; and a trigger assembly comprising: a static member comprising a pivoting attachment point, a top attachment point and a bottom attachment point, the pivoting attachment point for pivoting attachment to the snow plow blade, and the top and bottom attachment points for releasable attachment to the intermediate structure; a pin assembly; and a biasing means attached, at one end, to the pin assembly and attached, at an other end, to the static member such that the trigger assembly is biased toward a default position in which the pin member and the static member are at their closest, wherein when applying a torque on the snow plow blade above a trigger threshold, the snow plow blade rotates from the default position to a triggered position.

(8) According to an aspect, the static member comprises a bottom end area and a top end area distal from the bottom end area, wherein the top attachment point is in the top end area and the bottom

attachment point and the pivoting attachment point are in the bottom end area.

(9) According to an aspect, the static member comprises an anchoring point in the bottom end area and to which the other end of the biasing means is attached.

(10) According to an aspect, the one end of the biasing means is attached to the pin assembly forward of and above the bottom attachment point.

(11) According to an aspect, the pin assembly comprises a pin member and a sleeve in which the pin member is releasably secured.

(12) According to an aspect, the pin assembly further comprises an eyebolt to which the one end of the biasing means is attached.

(13) According to an aspect, the eyebolt is adjustable in length relative to the pin member and thereby adjusts a force exerted by the biasing means.

(14) According to an aspect, the intermediate structure comprising an articulated assembly.

(15) According to an aspect, the static member has a distal end distant from the pivoting attachment, wherein the articulated assembly comprises a rear longitudinal blade, a front longitudinal blade pivotably mounted to the rear longitudinal blade, and a top link arm, and further wherein the front longitudinal blade is releasably attached to the pivoting attachment, the rear longitudinal blade is mounted to the interface bracket and the top link arm is releasably attached to the distal end of the static member and pivotably mounted to the interface bracket.

(16) According to an aspect, the top link arm is located above the front longitudinal blade.

(17) According to an embodiment, there is provided a trigger assembly for connecting a snow plow blade to an interface bracket, via an intermediate structure, the interface bracket for releasable connection to a vehicle, the trigger assembly comprising: a static member comprising a pivoting attachment point, a top attachment point and a bottom attachment point, the pivoting attachment point for pivoting attachment to the snow plow blade, and the top and bottom attachment points for releasable attachment to the intermediate structure; a pin assembly; and a biasing means attached, at one end, to the pin assembly and attached, at an other end, to the static member such that the trigger assembly is biased toward a default position in which the pin member and the static member are at their closest, wherein when applying a torque on the snow plow blade above a trigger threshold, the snow plow blade rotates from the default position to a triggered position.

(18) According to an aspect, the static member comprises a bottom end area and a top end area distal from the bottom end area, wherein the top attachment point is in the top end area and the bottom attachment point and the pivoting attachment point are in the bottom end area.

(19) According to an aspect, the static member comprises an anchoring point in the bottom end area and to which the other end of the biasing means is attached.

(20) According to an aspect, the one end of the biasing means is attached to the pin assembly forward of and above the bottom attachment point.

(21) According to an aspect, the pin assembly comprises a pin member and a sleeve in which the pin member is releasably secured.

(22) According to an aspect, the pin assembly further comprises an eyebolt to which the one end of the biasing means is attached.

(23) According to an aspect, the eyebolt is adjustable in length relative to the pin member and thereby adjusts a force exerted by the biasing means.

(24) According to an embodiment, there is provided an attachment assembly for attaching a snow plow blade to a vehicle, comprising: a pair of interface brackets adapted to be releasably secured to the vehicle; a pair of intermediate structures, wherein each one of the pair of intermediate structures comprises an articulated assembly comprising an axis in a transversal orientation, wherein the articulated assembly comprises a rear longitudinal blade, a front longitudinal blade pivotably mounted to the rear longitudinal blade about the axis, and a top link arm; and a transversal member joining the pair of intermediate structures in the transversal orientation.

(25) According to an aspect, the top link arm comprises a top face having a downward curvature.

(26) According to an aspect, the front longitudinal blade is releasably attached to the snow plow blade, the rear longitudinal blade is mounted to a respective one of the pair of interface brackets and the top link arm is releasably attached to snow plow blade and pivotably mounted to the rear longitudinal blade.

(27) Features and advantages of the subject matter hereof will become more apparent in light of the following detailed description of selected embodiments, as illustrated in the accompanying figures. As will be realized, the subject matter disclosed and claimed is capable of modifications in various respects, all without departing from the scope of the claims. Accordingly, the drawings and the description are to be regarded as illustrative in nature and not as restrictive and the full scope of the subject matter is set forth in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further features and advantages of the present disclosure will become apparent from the following detailed description, taken in combination with the appended drawings, in which:

(2) FIG. 1 is a perspective rear view of the attachment assembly with a snow plow blade mounted thereto in accordance with an embodiment;

(3) FIG. 2 is a perspective front view of the attachment assembly of FIG. 1 without a snow plow blade mounted thereto;

(4) FIG. 3 is a perspective view of the trigger assembly in accordance with an embodiment;

(5) FIG. 4 is a bottom view of the trigger assembly of FIG. 3;

(6) FIG. 5 is a picture depicting an example of a skid steer loader operable with the attachment assembly of FIGS. 1 and 2;

(7) FIG. 6 is a picture of the front view of a skid steer loader with a snow plow blade assembly mounted thereto, wherein the picture depicts the position of a snow plow blade mounted to the attachment assembly of FIGS. 1 and 2 with the snow plow blade shown supported on a single side simulating an uneven ground surface;

(8) FIG. 7 is a picture of the side of the attachment assembly installed on a skid steer loader illustrating the articulated assemblies operating independently from each other;

(9) FIG. 8 is a picture of the side of the attachment assembly installed on a skid steer loader illustrating the trigger assembly in the default position;

(10) FIG. 9 is a picture of the side of the attachment assembly installed on a skid steer loader illustrating the trigger assembly in the triggered position;

(11) FIGS. 10A and 10B are side views of a portion of the attachment assembly depicting the interaction between the front blade and the second transversal member, wherein FIG. 10A depicts both of the front blades in intermediary positions while FIG. 10B depicts the frontmost front blade in an elevated blade position and the rearmost front blade in a lowered blade position; and

(12) FIG. 11 is a picture of the side of an attachment assembly of the PRIOR ART.

(13) It will be noted that throughout the appended drawings, like features are identified by like reference numerals.

DETAILED DESCRIPTION

(14) The realizations will now be described more fully hereinafter with reference to the accompanying figures, in which realizations are illustrated. The foregoing may, however, be embodied in many different forms and should not be construed as limited to the illustrated realizations set forth herein.

(15) With respect to the present description, references to items in the singular should be understood to include items in the plural, and vice versa, unless explicitly stated otherwise or clear from the text. Grammatical conjunctions are intended to express any and all disjunctive and

conjunctive combinations of conjoined clauses, sentences, words, and the like, unless otherwise stated or clear from the context. Thus, the term “or” should generally be understood to mean “and/or” and so forth.

(16) Recitation of ranges of values and of values herein or on the drawings are not intended to be limiting, referring instead individually to any and all values falling within the range, unless otherwise indicated herein, and each separate value within such a range is incorporated into the specification as if it were individually recited herein. The words “about”, “approximately”, or the like, when accompanying a numerical value, are to be construed as indicating a deviation as would be appreciated by one of ordinary skill in the art to operate satisfactorily for an intended purpose. Ranges of values and/or numeric values are provided herein as examples only, and do not constitute a limitation on the scope of the described realizations. The use of any and all examples, or exemplary language (“e.g.,” “such as”, or the like) provided herein, is intended merely to better illuminate the exemplary realizations and does not pose a limitation on the scope of the realizations. No language in the specification should be construed as indicating any unclaimed element as essential to the practice of the realizations. The use of the term “substantially” is intended to mean “for the most part” or “essentially” depending on the context. It is to be construed as indicating that some deviation from the word it qualifies is acceptable as would be appreciated by one of ordinary skill in the art to operate satisfactorily for the intended purpose.

(17) In the following description, it is understood that terms such as “first”, “second”, “top”, “bottom”, “above”, “below”, and the like, are words of convenience and are not to be construed as limiting terms.

(18) The terms “top”, “up”, “upper”, “bottom”, “lower”, “down”, “vertical”, “horizontal”, “interior” and “exterior” and the like are intended to be construed in their normal meaning in relation with normal installation of the product, with indication of “front” referring to the extremity of the vehicle on which the snow plow blade assembly is mounted to, “longitudinal” to the general orientation rear-front of the vehicle, and “transversal” to the orientation perpendicular to longitudinal along an horizontal plane. indication of “down” refers to the direction toward the ground.

(19) It should further be noted that for purposes of this disclosure, the term “coupled” means the joining of two members directly or indirectly to one another. Such joining may be stationary in nature or movable in nature and/or such joining may allow for the flow of fluids, electricity, electrical signals, or other types of signals or communication between two members. Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another. Such joining may be permanent in nature or alternatively may be removable or releasable in nature.

(20) In realizations, there are disclosed components of a snow plow and more particularly of a snow plow blade assembly, aka blade assembly.

(21) It will be noted that throughout the appended drawings, like features are identified by like reference numerals.

(22) Referring now to the drawings, and more particularly to FIGS. **1** and **2**, a snow plow blade assembly **100** is adapted to be mounted to a snow plow (not depicted) of the type of a skid steer loader **90** (see FIG. **5**) or another vehicle of similar attachment configuration.

(23) The configuration of the skid steer loader **90** is adapted for the operator to pass in front and between the forks / hydraulic arms to enter the driving cabin **92**. Further, the skid steer loader **90** is a polyvalent vehicle that must be able to perform multiple operations, and thus be able to operate with a variety of tools, including a snow plow blade assembly **100**. Thus, the present snow plow blade assembly **100** must both be able to be easily mounted to and dismounted from the skid steer loader **90** and further must provide the clearance for the operator of the skid steer loader **90** to

relatively easily enter and exit the cabin **92** without having to dismount the snow plow blade assembly **100** from the skid steer loader **90**.

(24) The snow plow blade assembly **100** comprises on its rear wall **112** a pair of holder assemblies **150** adapted for the attachment of the blade **110** to the attachment assembly **200** itself mounted to the skid steer loader **90**.

(25) The snow plow blade assembly **100** comprises a central section **120** and, on each side, a sidewall **130** mounted to the central section **120**.

(26) The snow plow blade assembly **100** further comprises one or more hydraulic jacks **160** that may be adapted to control configurations of snow plow blade assembly **100**, such as the width of the snow plow blade assembly **100**, the position of the sidewalls **130** relative to the central section **120**, etc., with the hydraulic jacks **160** being connected to the hydraulic power of the skid steer loader **90**.

(27) Referring particularly to FIGS. **1** and **2**, the attachment assembly **200** comprises a pair of interface brackets **210** adapted to interface with the fork interfaces **94** of the skid steer loader **90**. The interface brackets **210** comprise on their rear face **212** interfacing walls **220** contacting the fork interfaces **94** such as the interface brackets **210** and the fork interfaces **94** are secured to each other, allowing to operate the snow plow blade assembly **100**, which comprises pushing, pulling, lifting and lowering the snow plow blade assembly **100** without the interface between the skid steer loader **90** and the snow plow blade assembly **100** being released.

(28) The interface brackets **210** comprise an interface wall **220** and a side wall **222** extending rearward from the interface walls, wherein the interface walls **220** define an interface plane limited transversally by the side walls **222**. The interface brackets **210** also comprise a top wall **224** extending rearward from the interface wall **220**.

(29) The attachment assembly **200** consists of a single rigid structure adapted to interface with the skid steer loader **90** that remains whole when either or both of a) the attachment assembly **200** is dismounted from the skid steer loader **90** or b) no snow plow blade **110** is mounted to the attachment assembly **200**.

(30) The attachment assembly **200** comprises, on each side, mounted to the front face **214** of the respective interface bracket **210**, an intermediate structure **125**. In the described realization, both intermediate structures **125** comprise an articulated assembly **230**.

(31) The attachment assembly **200** comprises a first transversal member **232** joining the intermediate structures **125** about the interface brackets **210**. According to the depicted realization, the first transversal member **232** is also mounted to the interface brackets **210** for increased rigidity.

(32) The first transversal member **232** is typically made of a hollow beam providing a channel for hydraulic hoses (see FIG. **7** for example) to travel therethrough.

(33) The two articulated assemblies **230** are further joined by a second transversal member **234** providing the desired rigidity to the attachment assembly **200**.

(34) According to a realization, the attachment assembly **200** comprises, mounted to the top of the second transversal member **234**, a step member **260** featuring a non-slippery top face **262** adapted for the operator of the skid steer loader **90** to step on when entering or exiting the cabin **92** of the skid steer loader **90**.

(35) The articulated assemblies **230** comprise at their front a low mounting component **242** adapted for the lower portion **152** of a respective holder assembly **150** to be attached thereto.

(36) The articulated assemblies **230** further comprise a top link arm **250** which is mounted distant from the front longitudinal blade **274** and the rear longitudinal blade **272**. Each one of the top link arms **250** is attached at a rear junction **252** to the front face **214** of a respective interface bracket **210** and at its front junction **254** to the top portion **154** of the respective holder assembly **150**. At least one, but preferably both attachments of the top link arms **250** are pivotably mounted to their respective mounting components. The top link arms **250** provide a fixed-length attachment between the front face **214** of the interface brackets **210** and the rear wall **112** of the blade **110** while

allowing modification of the angle between the interface brackets **210** and the snow plow blade **110**.

(37) The top link arms **250** feature a top face **256** having a downward curvature (e.g., a U-shape link arm). The downward curvature of the top face **256** results in each of the top link arm **250** extending above and about their respective articulated assembly **230**, and thereby providing clearance over the top face **256** for the operator of the skid steer loader **90** to easily travel in and out of (i.e., enter and exit) the space delimited by the attachment assembly **200** with limited efforts; the operator having no surface to climb and the components to step over being kept closer to the ground.

(38) According to a realization, the top face **256** of the top link arms **250** feature a U-shape, extending downward close to the front face **214** of the interface brackets **210** and to the rear wall **112** of the blade **110**, and extending about horizontal in-between above the articulated assemblies **230**.

(39) Each one of the articulated assemblies **230** comprises a rear blade **272** (aka rear longitudinal blade) and a front blade **274** (aka front longitudinal blade) extending longitudinally from the front face **214** of the interface bracket **210** to the low portion of the holder assembly **150**, with the front blade **274** being pivotably mounted to the holder assembly **150**.

(40) The rear blade **272** and the front blade **274** are pivotably mounted to each other at an axis **265**, participating in a first articulation **275** between the interface brackets **210** and the blade **110**. The pivot mountings of the top link arms **250** provide a second articulation **285** between the interface brackets **210** and the blade **110**.

(41) Referring additionally to FIGS. **6** and **7**, one must note that the articulations **275**, **285** of the attachment assembly **200** on the left side operate independently from the articulations **275**, **285** of the attachment assembly **200** on the right side. Accordingly, in response to unevenness of the ground, the articulations **275**, **285** on one side (e.g., the left side) of the snow plow blade assembly **100** allows the left side of the snow plow blade **110** to move up or down without the right side of the snow plow blade **110** moving up or down; the position of the right side of the snow plow blade **110** is subject to the straight configuration of the snow plow blade **110** and of the configuration of its ground-contacting edge **116** and not of a displacement forced by the articulations **275**, **285** on the right side.

(42) Referring additionally to FIGS. **10A** and **10B**, the front blade **274** comprises a top limit face **284** and bottom limit face **288** extending rearward from the pivoting attachment of the front blade **274** to the rear blade **272**. The limit faces **284** and **288** extend about the second transversal member **234**. The top limit face **284** is facing substantially downward while the bottom limit face **288** is facing substantially upward. The first limit face and the second limit face have a clearance in-between where is housed at least a portion of the second transversal member **234** acting as a rotation limit component cooperating with the top jaw and the bottom jaw. When the blades **272**, **274** rotate relative to each other, one of the top limit face **284** and the bottom limit face **288** approaches the second transversal member **234** and, at the limit of the displacement of the snow plow blade **110** relative to its mounting to the skid steer loader **90**, abuts the second transversal member **234**, thereby limiting the displacement of the snow plow blade **110** within the limits of the articulated assembly **230**.

(43) According to a realization, the second transversal member **234** features a pair of sloped front faces **236**, **238**, providing maximum range for pivoting of the articulation **275** requiring a minimal clearance between the top limit face **284** and the bottom limit face **288** and a minimal distance between the axis **265** and the second transversal member **234**.

(44) FIGS. **10A** and **10B** are side views depicting the interaction between the front blade **274** and the sloped faces **236** and **238** of the second transversal member **234**, with the rear blade **272** not depicted for the sake of clarity. FIG. **10A** depicts when both of the frontmost front blade **274'** and the rearmost front blade **274''** (hidden behind) are in intermediary positions, aka when the top limit

face **284** and the bottom limit face **288** are not contacting the sloped faces **236** and **238**. FIG. **10B** depicts the frontmost front blade **274'** in an elevated blade position and the rearmost front blade **274''** in a lowered blade position. In the elevated blade position, the contact between the sloped face **236** and the limit face **284** limits further pivoting of the rear blade **272** and of the front blade **274** relative to each other. In the lowered blade position (partially hidden), the contact between the sloped face **238** and the limit face **288** limits further pivoting of the rear blade **272** and the front blade **274''** relative to each other.

(45) It is to be noted, as depicted, that the pivoting of the left and of the right front blade/rear blade assemblies are independent from each other.

(46) According to realizations, the top and bottom limit faces **284**, **288** abuts at the limit of the articulation **275** either one of the sloped faces **236**, **238** or one of the apexes **246**, **248**, i.e., top and lowest portions, of the second transversal member **234**.

(47) According to a realization, the limit faces **284**, **288** consist in a single arched face comprising a top component and an abutting component capable of abutting faces of the second transversal member **234**.

(48) According to a realization, the angle between the sloped faces **236**, **238** is of about ninety (90) degrees, with the angles being about symmetrical (forty-five (45) degrees above and forty-five (45) degrees below) relative to a plane about the horizontal when the snow plow blade **110** is contacting horizontal ground.

(49) Referring for comparison to the assembly of the PRIOR ART depicted on FIG. **11**, the attachment assembly of the PRIOR ART **500** must be mounted to multiple points to the vehicle, more precisely comprising a first mounting component **510** mounted to the bottom of the vehicle arm and a second mounting component **520** mounted to about the middle of the vehicle arm. The attachment assembly of the PRIOR ART **500** a) does not feature a simple and rigid structure that is easy to mount to the vehicle, and b) does not provide clearance for the operator to enter in the space delimited between the snow plow blade **110** and the vehicle without much efforts since it requires the operator to climb over the highest arm **530** or to pass over the snow plow blade **110** to reach this delimited space. Numerous additional advantages result from the hereindescribed attachment assembly **200** when compared to the attachment assembly of the PRIOR ART **500**.

(50) Referring now additionally to FIGS. **3**, **4**, **8** and **9**, the snow plow blade assembly **100**, in some realizations, comprises a trigger assembly **300** capable of interfacing on the rear side with the intermediate structure **125** and on the front side with the holder assemblies **150** of the snow plow blade **110**. The trigger assembly **300** is adapted for the blade **110**, upon hitting a non-moveable object about the ground, thus upon undergoing a torque above a trigger threshold, to have the snow plow blade **110** moving from a default position (see FIG. **8**) to a triggered position (see FIG. **9**) through which the stopping of the vehicle (e.g., skid steer loader **90**) from its forward motion is dampened by the displacement of the snow plow blade **110**.

(51) One should note that the trigger assembly **300** comprises a similar attachment interface on its rear to the holder assembly **150** and a similar attachment interface on its front to the attachment assembly **200**, thereby allowing to have realizations with trigger assemblies **300** and realizations free of trigger assemblies **300** comprising the same attachment assembly **200** and snow plow blade **110**.

(52) The trigger assembly **300** comprises a static member comprising a nested arm **310**, an external arm **320**, a pivoting attachment point **330** for rotative attachment to the snow plow blade. The trigger assembly **300** further comprises biasing means **340** (e.g., extendable springs **340**).

(53) It should be noted that, according to an embodiment, the external arm **320** has top end attachment point **336** in a top end area distant from the pivoting attachment point **330**. Still according to this embodiment, the front longitudinal blade **274** is releasably attached to the bottom attachment point **332**, the rear longitudinal blade **272** is mounted to the interface bracket **210** and the top link arm **250** is releasably attached to the top end attachment point **336** of the external arm

320 and pivotably mounted to the interface bracket **210**.

(54) The trigger assembly **300** comprises a pivoting attachment point **330** in a bottom area of the static member which is releasably attached at the bottom to the snow plow blade **110**. According to realizations, the nested arm **310** is mounted to/interfaced with the snow plow blade **110** at a first height and the external arm **320** is mounted to/interfaced to the snow plow blade **110** at a second height greater than the first height.

(55) The trigger assembly **300** comprises at the bottom end area an extension plate **342** extending rearward and comprising apertures (aka an anchoring point **344**) rearward to the pivoting attachment point **330** for attachment of the bottom extremity of the extendable springs **340**.

(56) It is to be noted that, in its default configuration, the extendable spring **340** is mounted to the external arm **320** rearward to the pivoting attachment point **330** and is mounted to the pin assembly in front and higher than where it is attached to the external arm **320** (e.g., at the anchoring point **344**). This configuration provides the tension necessary for the trigger assembly **300** to return to the default configuration upon release of the external torque.

(57) One should note that the external arm **320** and the extension plate **342** are static relative to each other. Another realization (see FIG. 7 for example) comprises a single static member performing both functions of the assembly comprising the depicted external arm **320** and extension plate **342**.

(58) The extendable springs **340** are further attached at their top to the pin assembly comprising the pin member **312** and the sleeve **334**. The pin assembly extends horizontally on both sides of the central portion static member. FIG. 3 further shows eyebolts **314** mounted to the pin member **312**. The pin assembly is releasably attached to the holder assembly **150** to move concurrently with the blade **110**. For clarity, the pin assembly is free to move relative to the nested arm **310** and the external arm **320**; i.e., it is not attached to the static member.

(59) Adjusting the length of the portion of the eyebolts **314** extending downward allows to adjust the force exerted by the extendable springs **340** between the eyebolts **314** and the anchoring point **344**, thus the sensitivity of the trigger assembly **300**.

(60) The external arm **320** comprises two parallel plates **322**, **324** providing a clearance therebetween for housing the nested arm **310**.

(61) It is worth noting that the pin member **312** is located frontward from top attachment point **336** of the external arm **320**.

(62) The trigger assembly **300** enters in a triggered position when a torque beyond a trigger threshold is applied to the pivoting attachment point **330** when the ground-contacting edge **116** of the blade **110** hits an obstacle. When triggered, the ground-contacting edge **116** tends to stay in place while the top edge **118** tends by inertia to continue its course forward. This results in the pin assembly rotating frontward (at its attachment point with the snow plow blade) relative to the pivoting attachment point **330**, thereby being moving away from the external arm **320**. In order for rotation to occur, the torque must exceed the force provided by the extendable springs **340** that extend further as rotation occurs.

(63) Accordingly, as soon as the torque applied to the blade **110** is released or decreased under a certain level, the pin assembly, under the forces exerted by the extendable springs **340**, tends to return to its default position, at its closest to the static member.

(64) According to realizations, the number of spring-loaded trigger assemblies **300** may be of a single one, two or more.

(65) According to realizations, the number, distribution and characteristics of the extendable spring(s) **340** may vary based on the desired sensitivity of the trigger assemblies **300**, the extension designed to be undergone by the extendable spring(s) **340** when reaching the triggered position, etc.

(66) According to realizations, the interfacing of components of the present snow plow blade assembly **100** comprises either releasable attachments (e.g., bolts and nuts, driven attachments such as hydraulic locks) or a combination of nestable components and releasable attachments ensuring

that the nesting components remain nested until the releasable attachments are removed from the locking position.

(67) While preferred embodiments have been described above and illustrated in the accompanying drawings, it will be evident to those skilled in the art that modifications may be made without departing from this disclosure. Such modifications are considered as possible variants comprised in the scope of the disclosure.

Claims

1. An attachment assembly for attaching a snow plow blade to a vehicle, comprising: an interface bracket adapted for releasable connection with the vehicle; an intermediate structure mounted to the interface bracket; and a trigger assembly comprising: a static member comprising a pivoting attachment point, a top attachment point and a bottom attachment point, the pivoting attachment point for pivoting attachment to the snow plow blade, and the top and bottom attachment points for releasable attachment to the intermediate structure; a pin assembly; and a biasing means attached, at one end, to the pin assembly and attached, at an other end, to the static member such that the trigger assembly is biased toward a default position in which the pin assembly and the static member are at their closest, wherein when applying a torque on the snow plow blade above a trigger threshold, the snow plow blade rotates from the default position to a triggered position, wherein the intermediate structure comprising an articulated assembly, wherein the static member has a distal end distant from the pivoting attachment point, and wherein the articulated assembly comprises a rear longitudinal bracket, a front longitudinal bracket pivotably mounted to the rear longitudinal bracket, and a top link arm, and further wherein the front longitudinal bracket is releasably attached to the pivoting attachment point, the rear longitudinal bracket is mounted to the interface bracket and the top link arm is releaseably attached to the distal end of the static member and pivotably mounted to the interface bracket.
2. The attachment assembly of claim 1, wherein the static member comprises a bottom end area and a top end area distal from the bottom end area, wherein the top attachment point is in the top end area and the bottom attachment point and the pivoting attachment point are in the bottom end area.
3. The attachment assembly of claim 2, wherein the static member comprises an anchoring point in the bottom end area and to which the other end of the biasing means is attached.
4. The attachment assembly of claim 3, wherein the one end of the biasing means is attached to the pin assembly forward of and above the bottom attachment point.
5. The attachment assembly of claim 1, wherein the pin assembly comprises a pin member and a sleeve in which the pin member is releasably secured.
6. The attachment assembly of claim 5, wherein the pin assembly further comprises an eyebolt to which the one end of the biasing means is attached.
7. The attachment assembly of claim 6, wherein the eyebolt is adjustable in length relative to the pin member and thereby adjusts a force exerted by the biasing means.
8. The attachment assembly of claim 1, wherein the top link arm is located above the front longitudinal bracket.
9. A trigger assembly for connecting a snow plow blade to an interface bracket, via an intermediate structure, the interface bracket for releasable connection to a vehicle, the trigger assembly comprising: a static member comprising a pivoting attachment point, a top attachment point and a bottom attachment point, the pivoting attachment point for pivoting attachment to the snow plow blade, and the top and bottom attachment points for releasable attachment to the intermediate structure; a pin assembly; and a biasing means attached, at one end, to the pin assembly and attached, at an other end, to the static member such that the trigger assembly is biased toward a default position in which the pin assembly and the static member are at their closest, wherein when applying a torque on the snow plow blade above a trigger threshold, the snow plow blade rotates

from the default position to a triggered position, wherein the intermediate structure comprises an articulate assembly, wherein the static member has a distal end distant from the pivoting attachment point, and wherein the articulated assembly comprises a rear longitudinal bracket, a front longitudinal bracket pivotably mounted to the rear longitudinal bracket, and a top link arm, and further wherein the front longitudinal bracket is releasably attached to the pivoting attachment point, the rear longitudinal bracket is mounted to the interface bracket and the top link arm is releasably attached to the distal end of the static member and pivotably mounted to the interface bracket.

10. The trigger assembly of claim 9, wherein the static member comprises a bottom end area and a top end area distal from the bottom end area, wherein the top attachment point is in the top end area and the bottom attachment point and the pivoting attachment point are in the bottom end area.

11. The trigger assembly of claim 10, wherein the static member comprises an anchoring point in the bottom end area and to which the other end of the biasing means is attached.

12. The trigger assembly of claim 11, wherein the one end of the biasing means is attached to the pin assembly forward of and above the bottom attachment point.

13. The trigger assembly of claim 9, wherein the pin assembly comprises a pin member and a sleeve in which the pin member is releasably secured.

14. The trigger assembly of claim 13, wherein the pin assembly further comprises an eyebolt to which the one end of the biasing means is attached.

15. The trigger assembly of claim 14, wherein the eyebolt is adjustable in length relative to the pin member and thereby adjusts a force exerted by the biasing means.
